AYE
TECH HUB

• SOLAR ENERGY • LESSON 02

Solar Radiation Basics
Reading the Sun

Irradiance, Peak Sun Hours & Tilt Optimisation

Lesson 2 of 30 | Solar Energy Program | Beginner

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- Solar irradiance — GHI, DNI, DHI and how they differ
- The solar constant and how atmosphere reduces it to 1,000 W/m²
- Peak Sun Hours (PSH) — the key to accurate solar system sizing
- How tilt angle, azimuth, and shading affect energy yield
- Using irradiance maps and PSH data for real project design

01 | What Is Solar Radiation?

Solar radiation is electromagnetic energy emitted by the sun across a broad spectrum of wavelengths — from ultraviolet through visible light to infrared. When this energy reaches a surface, it is called **solar irradiance**, measured in watts per square metre (W/m²). The total energy received over time is **solar irradiation** or **insolation**, measured in watt-hours per square metre (Wh/m² or kWh/m²).

For solar engineers, two numbers are used constantly: the **irradiance at a given moment** (how intense is the sun right now?) and the **daily irradiation** (how much solar energy fell on this surface today?). These drive every calculation from panel sizing to energy yield prediction.

THE SOLAR CONSTANT — Energy at the Edge of the Atmosphere

At the outer edge of Earth's atmosphere, solar irradiance averages 1,361 W/m² — this is called the Solar Constant. By the time sunlight reaches the ground, atmospheric absorption and scattering reduce this to roughly 1,000 W/m² on a clear day at sea level with the sun directly overhead. This 1,000 W/m² value is the Standard Test Condition (STC) used to rate every solar panel in the world.

02 | The Three Components of Solar Irradiance

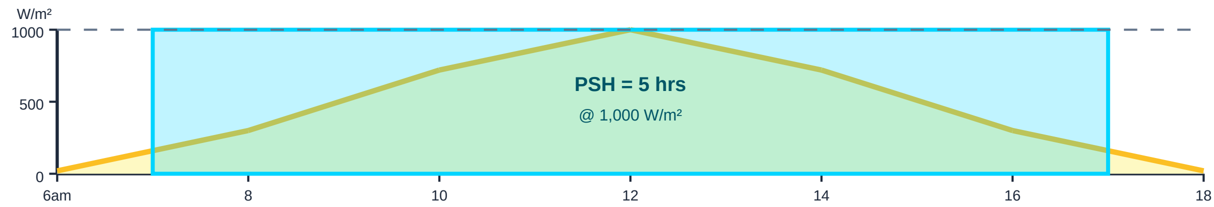
Solar irradiance reaching a horizontal surface is composed of three distinct components. Every solar resource assessment tool (PVsyst, PVGIS, SolarEdge) uses these components to calculate energy yield.

Component	Abbreviation	Description	Affects Which Systems?	How Measured
Global Horizontal Irradiance	GHI	Total solar radiation on a horizontal surface — the sum of DNI (projected) + DHI. The standard baseline resource value.	Fixed-tilt systems, flat rooftop, any horizontal surface	Pyranometer (thermopile or silicon)
Direct Normal Irradiance	DNI	Solar radiation arriving directly from the sun on a surface perpendicular to the sun's rays — the "beam" component only.	Concentrating solar (CSP), solar trackers, dual-axis tracking systems	Pyrheliometer on a sun tracker
Diffuse Horizontal Irradiance	DHI	Scattered sunlight from the sky dome — the indirect component. Important on overcast days or in humid climates.	All PV systems — DHI is always collected even when cloudy	Pyranometer with shadow band or disc
Plane of Array Irradiance	POA	The actual irradiance falling on the tilted panel surface. POA = DNI × cos(angle of incidence) + DHI + reflected. Used for energy yield calculation.	All PV design calculations — this is what actually hits the panel	Irradiance sensor at same tilt as panel

03 | Peak Sun Hours — The Key to Solar System Sizing

Peak Sun Hours (PSH) is the single most important concept in solar system sizing. It represents the number of hours per day that solar irradiance averages 1,000 W/m² — the equivalent full-sun hours. Because actual irradiance varies throughout the day (from near zero at dawn to ~1,000 W/m² at noon), PSH converts the real daily solar energy into a simple, usable number.

Peak Sun Hours — Equivalent Full-Sun Energy Concept



How to use PSH in system sizing:

Daily energy needed ÷ PSH = required panel capacity (kWp). Example: A home needs 10 kWh/day and the site has 5 PSH. Panel capacity needed = 10 ÷ 5 = 2 kWp (before losses).

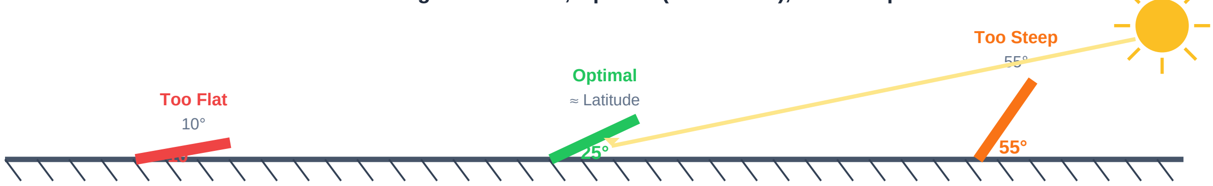
04 | PSH by Region — Global Solar Resource

Region / Country	Avg PSH (hrs/day)	Annual Irradiation (kWh/m²/yr)	Solar Potential
Sahara Desert (N. Africa)	7.0 – 9.0	2,500 – 3,300	World's highest — basis of large utility-scale solar farms
East Africa (Ethiopia, Eritrea, Kenya)	5.5 – 7.0	2,000 – 2,500	Excellent resource — ideal for off-grid and grid-tied projects
Middle East (UAE, Saudi Arabia)	6.0 – 8.0	2,200 – 2,900	Very high — major solar investment region
South Asia (India, Pakistan)	5.0 – 6.5	1,800 – 2,400	High — world's fastest growing solar markets
Southern Europe (Spain, Italy, Greece)	4.5 – 6.0	1,600 – 2,100	Good — EU solar leaders
Northern Europe (UK, Germany)	2.5 – 3.5	900 – 1,250	Lower but viable — significant installed capacity due to policy
South America (Chile, Brazil)	5.0 – 7.5	1,800 – 2,700	High — Atacama Desert is among world's best solar sites
Southeast Asia (Thailand, Vietnam)	4.5 – 5.5	1,600 – 2,000	Good — rapidly growing solar deployment

05 | Tilt Angle & Azimuth — Pointing Panels at the Sun

A solar panel generates maximum power when sunlight strikes it at 90° — perpendicular to the panel surface. Since the sun moves across the sky, the **tilt angle** (inclination from horizontal) and **azimuth** (compass direction the panel faces) are critical design parameters that determine annual energy yield.

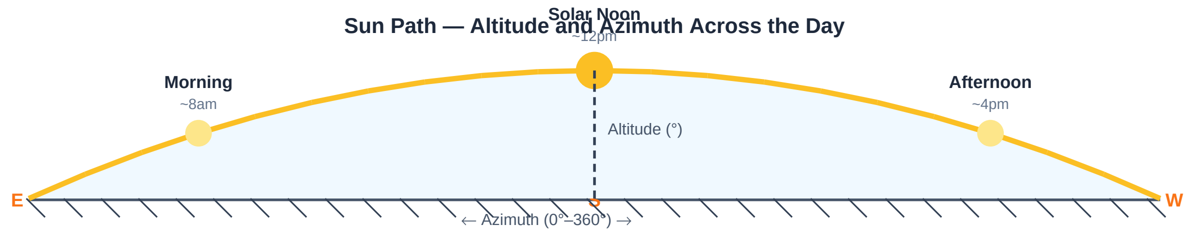
Panel Tilt Angle — Too Flat, Optimal (≈ Latitude°), Too Steep



Parameter	Definition	Optimal Value	Effect of Wrong Value
Tilt Angle (β)	Angle of panel surface from horizontal (0° = flat, 90° = vertical)	Equal to site latitude (±5°) for fixed systems in most cases	Each 10° from optimal can reduce annual yield by 2–8% depending on latitude
Azimuth (γ)	Compass direction the panel faces (0° = North in southern hemisphere, 180° = South in northern hemisphere)	True South (northern hemisphere) or True North (southern hemisphere)	East or West-facing panels lose 10–20% annual yield vs South-facing
Seasonal Tilt Adjustment	Changing tilt between summer and winter	Summer: latitude – 15° | Winter: latitude + 15°	Adjusting seasonally can increase annual yield by 5–10% vs fixed tilt
Dual-Axis Tracking	Panel automatically follows sun in azimuth and elevation	Always perpendicular to sun (maximum POA irradiance)	Increases energy yield by 25–40% vs fixed tilt — at higher cost and maintenance

06 | The Sun Path — Altitude and Azimuth Across the Sky

The sun moves across the sky in a predictable arc that changes with the time of day, season, and geographic latitude. Understanding this arc — called the **sun path** — is essential for panel placement, shading analysis, and designing solar tracking systems. Two angles define the sun's position: **altitude** (height above horizon) and **azimuth** (compass bearing).



Sun Position Concept	Definition	Engineering Use
Solar Altitude Angle (α)	Angle of sun above the horizon (0° at sunrise/sunset, 90° = directly overhead)	Determines panel tilt, shading geometry, and maximum irradiance
Solar Azimuth Angle (γ■)	Compass bearing of the sun from North (0°–360°)	Determines panel orientation and which objects cast shadows at which times
Solar Noon	The moment the sun reaches its highest altitude for the day — crosses the meridian	Peak irradiance time; panels should face solar noon direction for maximum yield
Hour Angle (ω)	Angle equivalent to time from solar noon (15° per hour)	Used in solar position equations (e.g., Declination + Hour Angle → Altitude)
Declination Angle (δ)	Angle between Earth's equatorial plane and the Earth–Sun line (±23.45°)	Changes with season; drives why optimal tilt and PSH vary through the year
Air Mass (AM)	Relative thickness of atmosphere sunlight passes through. AM1.0 = sun overhead; AM1.5 = standard test condition	AM1.5 spectrum is the standard for rating all PV panels and modules

07 | Shading, Losses & Real-World Yield Calculation

Loss Factor	Typical Value	Cause & Engineering Response
Shading losses	0 – 15%	Trees, buildings, chimneys, other panels. Use sun path analysis + PVsyst shading tool. Orient away from obstructions.
Soiling (dust, dirt)	1 – 5%	Accumulation on panel surface. Worse in desert sites. Mitigate with cleaning schedule and anti-soiling coatings.
Temperature losses	3 – 8%	Panel efficiency drops ~0.35–0.4%/°C above 25°C STC. Keep air gap behind panels; use low temperature coefficient modules.
DC cable losses	1 – 2%	Resistance in DC wiring from panels to inverter. Size cables for < 1% voltage drop at peak current.
Inverter losses	2 – 5%	Inverter conversion efficiency (93–98%). Select high-efficiency inverter, optimise MPPT string voltage.
Mismatch losses	0.5 – 2%	Panels in same string with different orientations or partial shading. Use microinverters or DC optimisers.
Total system losses	15 – 25%	Apply as performance ratio (PR = 0.75–0.85). Energy yield = PSH × panel kWp × PR

SIZING FORMULA — From Load to Panel Capacity

Step 1: Calculate daily energy need (kWh/day). Step 2: Look up PSH for your site. Step 3: Apply system losses (PR ≈ 0.80). Panel kWp = Daily kWh ÷ (PSH × PR). Example: 10 kWh/day ÷ (5.5 PSH × 0.80) = 2.27 kWp → round up to 2.4 kWp.

Reflection:

"Look up the average PSH for your city using PVGIS (re.jrc.ec.europa.eu/pvgis) or SolarAtlas. Find the optimal tilt angle for your latitude. Calculate how many kWp of solar panels a building consuming 30 kWh/day would need in your location, assuming a system performance ratio of 0.80."

Action checklist:

Next up — SOLAR-003:
Solar System Components — panels, batteries, charge controllers, inverters, and wiring in depth.

Community & support:
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